

ECE 315 – Signals and Systems, Spring 2026

Final Exam

10:30AM–12:00PM, May 12

Exam duration: 90 minutes

INSTRUCTIONS

1. This is a **closed-book** examination.
2. The exam has **five regular questions** comprising **35 points**, plus one **5-point bonus question**.
3. Show all work for full credit; clearly correct expressions in equivalent forms are sufficient.
4. Students may use scientific calculators, but wireless communication is prohibited. Other electronic devices, including computers, phones, tablets, and digital notebooks, are not allowed.
5. The mathematical formulas below may be helpful.

$$e^{jx} = \cos(x) + j \sin(x)$$

$$\sin^2(x) = \frac{1}{2}(1 - \cos(2x))$$

$$\cos^2(x) = \frac{1}{2}(1 + \cos(2x))$$

$$\sin(x + y) = \sin(x) \cos(y) + \cos(x) \sin(y)$$

$$\cos(x + y) = \cos(x) \cos(y) - \sin(x) \sin(y)$$

$$\sin(x) \cos(y) = \frac{1}{2}(\sin(x - y) + \sin(x + y))$$

$$\sum_{n=0}^{N-1} \alpha^n = \frac{1 - \alpha^N}{1 - \alpha}, \quad \alpha \neq 1$$

$$\sum_{n=0}^{\infty} \alpha^n = \frac{1}{1 - \alpha}, \quad |\alpha| < 1.$$

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6. The Fourier representation properties below may be helpful.

$$e^{-at}u(t) \quad \longleftrightarrow \quad \frac{1}{a + j\omega}, \quad a > 0$$

$$te^{-at}u(t) \quad \longleftrightarrow \quad \frac{1}{(a + j\omega)^2}, \quad a > 0$$

$$a^n u[n] \quad \longleftrightarrow \quad \frac{1}{1 - ae^{-j\omega}}, \quad |a| < 1$$

$$(n + 1)a^n u[n] \quad \longleftrightarrow \quad \frac{1}{(1 - ae^{-j\omega})^2}, \quad |a| < 1$$

$$\frac{d}{dt}x(t) \quad \longleftrightarrow \quad j\omega X(j\omega)$$

$$x[n - n_0] \quad \longleftrightarrow \quad X(e^{j\omega})e^{-j\omega n_0}$$

Question 1

(7 points)

Consider a discrete-time signal from a sensor:

$$x[n] = \begin{cases} 2, & n = -1, \\ 1, & n = 0, \\ -1, & n = 2, \\ 0, & \text{otherwise.} \end{cases}$$

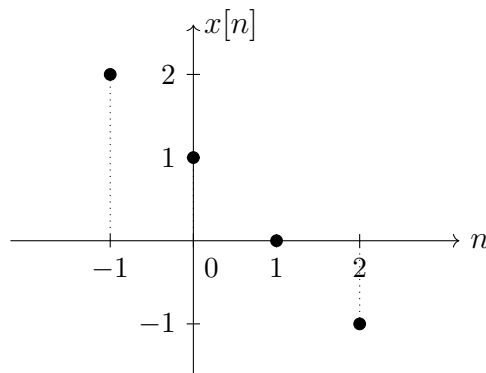


Figure 1: Discrete-time signal

- (i) **(2 pts)** Sketch the transformed signal $x[1 - n]$.
- (ii) **(2 pts)** Compute the energy and average power of $x[1 - n]$.

- (iii) **(3 pts)** A simple event detector compares the current sample with the sample one time step earlier:

$$y[n] = x[n]x[n-1].$$

Determine whether this detection system is memoryless, causal, linear, time-invariant, and invertible. Justify your answer.

Question 2**(8 points)**

Consider a continuous-time periodic square-wave signal:

$$x(t) = \begin{cases} 1, & 0 \leq t < 1, \\ -1, & 1 \leq t < 2, \end{cases}$$

and $x(t+2) = x(t)$ for all t .

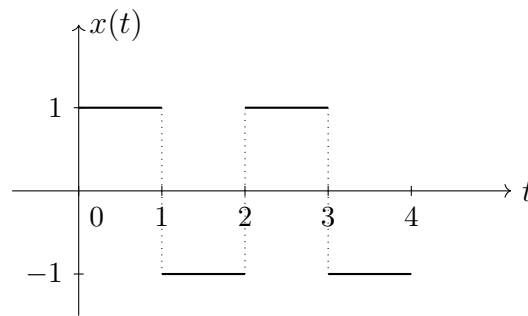


Figure 2: Square wave

- (i) **(1 pt)** Find the fundamental period T_0 and fundamental frequency ω_0 of $x(t)$.

(ii) **(4 pts)** Compute the continuous-time Fourier series coefficients of $x(t)$.

(iii) **(3 pts)** Pass the signal $x(t)$ through a low-pass filter with frequency response $H(j\omega)$ in Figure 3. Determine the Fourier series coefficients of the output $y(t)$.

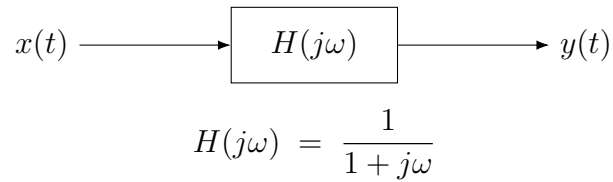


Figure 3: Low-pass filter

Question 3

(6 points)

Consider a positive rectangular pulse followed by a negative pulse:

$$x(t) = \begin{cases} 1, & |t| \leq 1, \\ -\frac{1}{2}, & 2 \leq t \leq 3, \\ 0, & \text{otherwise.} \end{cases}$$

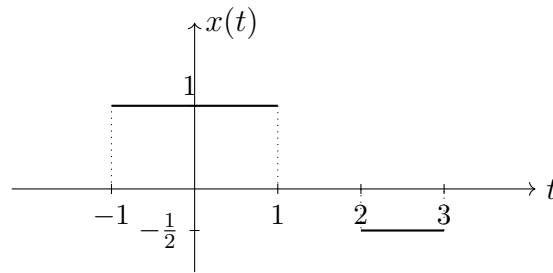


Figure 4: Rectangular pulse

- (i) **(3 pts)** Compute the continuous-time Fourier transform $X(j\omega)$ of $x(t)$.

(ii) **(1 pt)** Compute $X(j0)$ and explain its meaning using the figure of $x(t)$.

(iii) **(2 pts)** For the inverse Fourier transform representation of $x(t)$, select which types of convergence holds and justify your choice.

- ☐ Convergence everywhere
- ☐ Convergence with no energy difference
- ☐ Convergence except for discontinuities

Question 4**(7 points)**

Consider a causal linear time-invariant system with impulse response

$$h(t) = (e^{-t} - e^{-2t}) u(t).$$

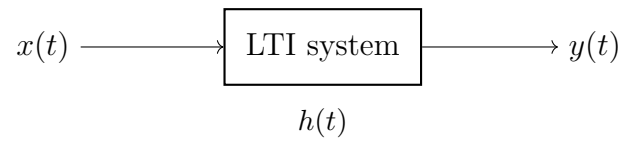


Figure 5: LTI system

- (i) **(2 pts)** Compute the frequency response $H(j\omega)$.

- (ii) **(2 pts)** Assuming zero initial conditions, find the differential equation relating input $x(t)$ and output $y(t)$.

- (iii) **(3 pts)** Compute the system response $y(t)$ for the input

$$x(t) = e^{-t}u(t).$$

Question 5

(7 points)

Consider a negative feedback system with input $x[n]$ and output $y[n]$, where the forward and feedback frequency responses are

$$H_1(e^{j\omega}) = \frac{1}{1 - \frac{1}{2}e^{-j\omega}} \quad \text{and} \quad H_2(e^{j\omega}) = \frac{1}{4}e^{-j\omega}.$$

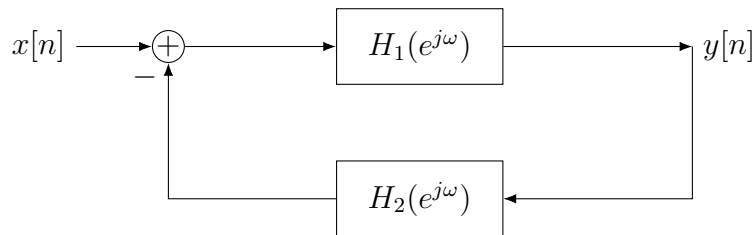


Figure 6: Negative feedback system

- (i) **(2 pts)** Determine the frequency response of the feedback system from $x[n]$ to $y[n]$.

- (ii) **(2 pts)** Find the difference equation relating input $x[n]$ and output $y[n]$.

- (iii) **(3 pts)** Compute the system response $y[n]$ for the input

$$x[n] = \left(\frac{1}{2}\right)^n u[n].$$

Question 6: Bonus

(5 bonus points)

Consider an ideal discrete-time band-pass filter,

$$H(e^{j\omega}) = \mathbf{1}_{\{|\omega| \leq \pi/2\}} - \mathbf{1}_{\{|\omega| \leq \pi/3\}} \quad \text{for } |\omega| \leq \pi.$$

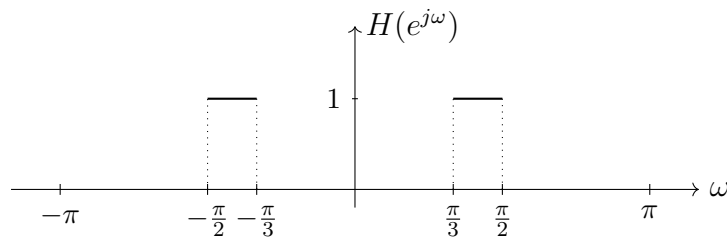


Figure 7: Band-pass filter

- (i) **(2 pts)** Determine the impulse response $h[n]$. You may use the identity

$$\frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} e^{j\omega n} d\omega = \begin{cases} \frac{\sin(\omega_c n)}{\pi n}, & n \neq 0, \\ \frac{\omega_c}{\pi}, & n = 0. \end{cases}$$

- (ii) (**2 pts**) Find the output $y[n]$ when the input $x[n]$ is given by

$$x[n] = 1 + 2 \cos\left(\frac{5\pi}{12}n\right).$$

- (iii) (**1 pt**) Is the ideal filter causal? Justify your answer.